

Too Long; Dye't Red: Doubling the Effective Number of Pages Using Stereoscopic 3D Printing

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Abstract—Academics, researchers, and other people who subject themselves to submitting manuscripts to journals and conferences with page limits often find themselves struggling to fit everything within the page limit. This leads to creative use of space and language, leaving out important figures, and leaving important proofs left as an exercise to the reader. This paper proposes a novel application of stereoscopic 3D effect to effectively double the amount of text that can fit on one page. We find that using this method, we can save on paper and ink, prevent back pain, improve the environment, and give new meaning to journals other than being websites with expensive PDFs.

Fig. 2. The subtractive colour wheel, showing what combinations of colours turn into which other colours. Please, imagine this, colours yourself.

I. INTRODUCTION

Page limits have existed since time immemorial. Early page limits were mostly physical, like when one would run out of clay to make another tablets [6]. This resulted in text needing to be very short and concise. After the invention of paper, and later the printing press [7], it became possible to pack far more words in the same amount of space. Bibles even optimised this by using extra thin paper [8].

Nowadays, most journals are digital [4], [1], and some journals are even digital only [3] (even though Elsevier, Springer, and others still like to charge you for them as if they were printed on parchment paper and handwritten by a monk using a quill). The tradition of having page limits has remained, however, even for journals that no longer do a print edition (unlike the clearly superior SIGBOMK which is still available in print).

Some ways to get more words on a page is by adjusting the page margins, reducing the font size, reducing vertical spacing, among others [2]. Whilst these do increase the amount of words one can fit on a page (thus reducing the number of pages necessary), it generally only saves one or two pages per submission.

Not having glasses – the journal could provide these 3D glasses have existed for over a century [5]. Even though different versions exist, one of the most known versions is using red and cyan glasses. Famous films such as *Bwana Devil*, *Kiss Me Kate*, *House of Wax*, *Hondo*, *Dial M for Murder*, and *Creature from the Black Lagoon* used this technology in the 1950s. One useful aspect of this method is that it can easily be used for any medium. Later 3D glasses use polarised filters, those who print papers, it should only be a small price to pay to make all figures black-and-white as well.

This paper outlines a novel application of stereoscopic 3D to reliably double the number of pages available by using anaglyphs.

V. ETHICAL CONSIDERATIONS

Outline of Section II we explain how and printing works in Section III we elaborate on the disadvantages of the method

considerations why they did not opt to use this method. Then again, that might make them write more and thus use more paper.

VI. CONCLUSION

Page limits are a big problem for academics worldwide. This paper presents a solution using stereoscopic 3D that solves that issue. We observe that there are many advantages to using this technique, such as doubling the amount of words available per manuscript, reducing back pain, avoiding overweight charges on air planes, saving trees, and reducing the use of black printer ink. Even though there are small disadvantages, we believe they are negligible and easily overcome. All in all, we conclude that there is a great advantage to using this technology and that it can be adopted right now.

Future Work

Future researchers might want to investigate the best way to do page ordering using this method.

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In Section IV we elaborate on the disadvantages (which is a short section, because there are few). In Section V we discuss the ethics of our work. Lastly, in Section VI we draw our conclusion.

II. HOW IT WORKS

We write our manuscript text as we normally would, taking into account to avoid the use of colour. We then combine sets of two pages on one piece of paper by printing one page exclusively using red (i.e. by combining yellow and magenta ink), and one page exclusively using cyan. Places where the colours overlap will turn black, as shown in Figure 2. The text can then be read using special glasses that have a red or cyan filter, blocking out the red or cyan part respectively, such as can be seen in Figure 1.

III. ADVANTAGES

This method offers many advantages – too many to name all of them, but here are a few:

- 1) Double the amount of words per page – one page counts as two, and when double-sided even as four, allowing you to put out more knowledge to the world;
- 2) Less back pain from carrying around have stacks of paper – when printing research the weight of papers adds